

CHAPTER II.2, WEEK 2

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2.1. Let A be a ring, $X = \operatorname{Spec} A$ its spectrum. Fix an element $f \in A$, $D(f) = V(f)^c \subset X$ the associated distinguished open set. Show that $(D(f), \mathcal{O}_X|_{D(f)}) \cong \operatorname{Spec} A_f$ as locally ringed spaces.

Proof. Let $Y = \operatorname{Spec} A_f$, $S = \{f^n\}_{n \geq 1}$, and $\phi : A \rightarrow S^{-1}A = A_f$ be the natural map $a \mapsto a/1$. Note that $f^n \in \mathfrak{p}$ prime for some $n \geq 1$ iff $f \in \mathfrak{p}$ —that is, $S \cap \mathfrak{p} = \emptyset$ iff $\mathfrak{p} \notin D(f)$. Thus, AM 3.11.iv gives us a bijection

$$\begin{aligned} D(f) &\longleftrightarrow Y \\ \mathfrak{p} &\longrightarrow S^{-1}\mathfrak{p}, \\ \phi^*(\mathfrak{q}) &\longleftarrow \mathfrak{q}. \end{aligned}$$

Clearly the map $\mathfrak{q} \mapsto \phi^*(\mathfrak{q})$ is continuous, so it suffices to show that $\mathfrak{p} \mapsto S^{-1}\mathfrak{p}$ is continuous.

Let $V(\mathfrak{b}) \subset A_f$ be closed, and denote $\psi : \mathfrak{p} \mapsto S^{-1}\mathfrak{p}$. AM 3.11.i gives us that $\mathfrak{b} = S^{-1}\mathfrak{a}$ is extended. Then $\mathfrak{p} \supset \mathfrak{a}$ implies $S^{-1}\mathfrak{p} \supset S^{-1}\mathfrak{a} = \mathfrak{b}$ whenever $f \notin \mathfrak{a}, \mathfrak{p}$, so that $V(\mathfrak{a}) \cap D(f) \subset \psi^{-1}(\mathfrak{b})$. Similarly, $S^{-1}\mathfrak{p} \supset S^{-1}\mathfrak{a}$ implies $\phi^*(S^{-1}\mathfrak{p}) \supset \phi^*(S^{-1}\mathfrak{a}) \supset \mathfrak{a}$. We conclude that $\psi^{-1}(V(\mathfrak{b})) = V(\mathfrak{a}) \cap D(f) \subset D(f)$ is closed, so that $\psi = (\psi^*)^{-1}$ is continuous.

Note that $(\mathcal{O}_X|_{D(f)})_{\mathfrak{p}} \cong A_{\mathfrak{p}} \cong (A_{\mathfrak{p}})_f \cong (A_f)_{\psi(\mathfrak{p})} \cong (\mathcal{O}_Y)_{\psi(\mathfrak{p})}$ for each $\mathfrak{p} \notin f$. Call this isomorphism $\alpha_{\mathfrak{p}} : (\mathcal{O}_X|_{D(f)})_{\mathfrak{p}} \rightarrow (\mathcal{O}_Y)_{\psi(\mathfrak{p})}$, and let

$$\alpha : \sqcup_{\mathfrak{p} \in D(f)} (\mathcal{O}_X|_{D(f)})_{\mathfrak{p}} \rightarrow \sqcup_{\mathfrak{q} \in Y} (\mathcal{O}_Y)_{\mathfrak{q}}$$

be defined by $\alpha|_{(\mathcal{O}_X|_{D(f)})_{\mathfrak{p}}} = \alpha_{\mathfrak{p}}$. Then we have a map

$$\begin{aligned} \phi^{\sharp}(U) : \mathcal{O}_X|_{D(f)}(U) &\rightarrow (\phi^*)_* \mathcal{O}_Y(U) \\ s &\mapsto \alpha \circ s \circ \phi^*. \end{aligned}$$

We have that

$$\alpha \circ s|_V \circ \phi^* = \alpha \circ s \circ \phi^*|_{(\phi^*)^{-1}(V)}$$

as functions on $(\phi^*)^{-1}(V)$ whenever $V \subset U \subset D(f)$ are open and $s \in \mathcal{O}_X(U)$, so $\phi^{\sharp}(U)$ gives a sheaf homomorphism $\phi^{\sharp} : \mathcal{O}_X|_{D(f)} \rightarrow \mathcal{O}_Y$. $\phi^{\sharp}_{\mathfrak{p}} = \alpha_{\mathfrak{p}}$ is an isomorphism of rings for all $\mathfrak{p} \in D(f)$, so $\phi^{\sharp}$ is itself a sheaf isomorphism. Then $(f, f^{\sharp})$ is an isomorphism of locally ringed spaces, since ring isomorphisms preserve maximality. $\square$

2.2. Let $(X, \mathcal{O}_X)$ be a scheme, and let $U \subset X$ be any open subset. Show that $(U, \mathcal{O}_X|_U)$ is a scheme. We call this the *induced scheme structure* on the open set U , and we refer to $(U, \mathcal{O}_X|_U)$ as the open subscheme of U .

Proof. Choose a basic cover $\{D(f_i)\}$ of U . Then each

$$(D(f_i), (\mathcal{O}_X|_U)_{D(f_i)}) = (D(f_i), \mathcal{O}_X|_{D(f_i)}) \cong \operatorname{Spec} A_{f_i}$$

by 2.1. We conclude that $\{D(f_i)\}$ is an affine open cover of $(U, \mathcal{O}_X|_U)$, so that the latter locally ringed space is a sheaf. $\square$

2.3. (*) A scheme $(X, \mathcal{O}_X)$ is *reduced* if for every open set $U \subset X$, the ring $\mathcal{O}_X(U)$ has no nilpotent elements.

(a) Show that $(X, \mathcal{O}_X)$ is reduced if and only if for every $p \in X$, the local ring $\mathcal{O}_{X,p}$ has no nilpotent elements.

(b) Let $(X, \mathcal{O}_X)$ be a scheme. Let $(\mathcal{O}_X)_{\text{red}}$ be the sheaf associated to the presheaf $U \mapsto \mathcal{O}_X(U)_{\text{red}}$, where for any ring A , we denote by A_{red} the quotient of A by its nilradical $\mathfrak{N}(A)$. Show that $(X, (\mathcal{O}_X)_{\text{red}})$ is a scheme. We call it the *reduced scheme* associated to X , and denote it by X_{red} . Show that there is a morphism of schemes $X_{\text{red}} \rightarrow X$, which is a homeomorphism on the underlying topological spaces.

(c) Let $f : X \rightarrow Y$ be a morphism of schemes, and assume that X is reduced. Show that there is a unique morphism $g : X \rightarrow Y_{\text{red}}$ such that f is obtained by composing g with the natural map $Y_{\text{red}} \rightarrow Y$.

2.4. Let A be a ring, and let $(X, \mathcal{O}_X)$ be a scheme. Given a morphism $f : X \rightarrow \text{Spec } A$, we have an associated map $f^\# : \mathcal{O}_{\text{Spec } A} \rightarrow f_* \mathcal{O}_X$ on sheaves. Taking Global sections, we obtain a homomorphism $A \rightarrow \Gamma(X, \mathcal{O}_X)$. Thus, there is a natural map

$$\alpha : \text{Hom}_{\mathfrak{S}\mathfrak{ch}}(X, \text{Spec } A) \rightarrow \text{Hom}_{\mathfrak{R}\mathfrak{ing}}(A, \Gamma(X, \mathcal{O}_X)).$$

Show that α is bijective.

Proof. We begin by showing the result for $X = \text{Spec } B$ affine. In this case, α has an inverse map given by

$$\begin{aligned} \beta : \text{Hom}_{\mathfrak{R}\mathfrak{ing}}(A, B) &\longrightarrow \text{Hom}_{\mathfrak{S}\mathfrak{ch}}(\text{Spec } B, \text{Spec } A) \\ \varphi &\longmapsto (f, f^\#), \end{aligned}$$

where

$$\begin{aligned} f(\mathfrak{p}) &= \varphi^{-1}(\mathfrak{p}), \\ f^\#(s)(\mathfrak{p}) &= \varphi_{f(\mathfrak{p})}(s(f(\mathfrak{p}))). \end{aligned}$$

It suffices to show that β and α indeed compose to the identity.

Fix $(f, f^\#) \in \text{Hom}_{\mathfrak{S}\mathfrak{ch}}(\text{Spec } B, \text{Spec } A)$, and denote

$$(\beta \circ \alpha)(f, f^\#) = (g, g^\#).$$

Let $\phi_{f(\mathfrak{p})} : A \rightarrow A_{f(\mathfrak{p})}$ and $\phi'_{\mathfrak{p}} : B \rightarrow B_{\mathfrak{p}}$ be the natural maps. The commutative square

$$\begin{array}{ccc} A & \xrightarrow{f^\#(\text{Spec } A)} & B \\ \phi_{f(\mathfrak{p})} \downarrow & & \downarrow \phi'_{\mathfrak{p}} \\ A_{f(\mathfrak{p})} & \xrightarrow{f^\#_{f(\mathfrak{p})}} & B_{\mathfrak{p}} \end{array}$$

gives us that

$$\begin{aligned}
g(\mathfrak{p}) &= (f^\sharp(\mathrm{Spec} A))^{-1}(\mathfrak{p}) \\
&= (\phi'_\mathfrak{p} \circ f^\sharp(X))^{-1}(m_\mathfrak{p}) \\
&= (f^\sharp_{f(\mathfrak{p})} \circ \phi_{f(\mathfrak{p})})^{-1}(m_\mathfrak{p}) \\
&= \phi_{f(\mathfrak{p})}^{-1}(m_{f(\mathfrak{p})}) \\
&= f(\mathfrak{p}),
\end{aligned}$$

since $f^\sharp$ is a local homomorphism. Then

$$\begin{aligned}
g^\sharp(s)(\mathfrak{p}) &= f^\sharp(X)_{g(\mathfrak{p})}(s(g(\mathfrak{p}))) \\
&= f^\sharp_{f(\mathfrak{p})}(s(f(\mathfrak{p}))) \\
&= f^\sharp(s)(\mathfrak{p}).
\end{aligned}$$

implies $(g, g^\sharp) = (f, f^\sharp)$.

Fix $\varphi : A \rightarrow B$, and denote $\beta(\varphi) = (f, f^\sharp)$, $(\alpha \circ \beta)(\varphi) = \psi$. Recall that $a \in A$ is identified with $\mathfrak{p} \mapsto a_\mathfrak{p} \in A_\mathfrak{p}$, and similarly for $b \in B$. Thus, $\psi(a) = \varphi(a)$ iff they're equal as maps $\mathrm{Spec} B \rightarrow \sqcup_{\mathfrak{p} \in \mathrm{Spec} B} B_\mathfrak{p}$. The commutative square

$$\begin{array}{ccc}
A & \xrightarrow{\varphi} & B \\
\downarrow & & \downarrow \\
A_{f(\mathfrak{p})} & \xrightarrow{\varphi_{f(\mathfrak{p})}} & B_\mathfrak{p}
\end{array}$$

gives us

$$\psi(a)(\mathfrak{p}) = \varphi_{f(\mathfrak{p})}(a(f(\mathfrak{p}))) = \varphi_{f(\mathfrak{p})}(a_{f(\mathfrak{p})}) = \varphi(a)(\mathfrak{p}).$$

Hence, $(\alpha \circ \beta)(\varphi) = \varphi$. We conclude that $\beta = \alpha^{-1}$, so α is a bijection.

Now, we consider the general case of X a scheme. Define a map

$$\begin{aligned}
\beta : \mathrm{Hom}_{\mathfrak{Ring}}(A, \Gamma(X, \mathcal{O}_X)) &\longrightarrow \mathrm{Hom}_{\mathfrak{Sch}}(X, \mathrm{Spec} A) \\
\varphi &\longmapsto (f, f^\sharp)
\end{aligned}$$

as follows. Fix an affine cover $\{U_i = \mathrm{Spec} B_i\}$ of X , and define $f : X \rightarrow \mathrm{Spec} A$ by $f(x) = \varphi^{-1}(\mathfrak{p}_x)$, where $\mathfrak{p}_x$ is the inverse image in $\Gamma(X, \mathcal{O}_X)$ of the unique maximal ideal $m_x \trianglelefteq (\mathcal{O}_X)_x$. Naturality of the restriction maps gives us that $f|_{U_i} = f_i$, where $(f_i, f_i^\sharp) = \beta(\rho_{U_i} \circ \varphi)$ is defined in the affine case.

Next, we define $f^\sharp$ as follows. Fix $V \subset X$ open and $s \in \mathcal{O}_{\mathrm{Spec} A}(V)$. Consider the sections $f_i^\sharp(V \cap U_i)(s) \in f_{i*}\mathcal{O}_X(V \cap U_i)$, with $f_i^\sharp$ as above. We have that

$$f_i^\sharp(V \cap U_i \cap U_j)(s)(\mathfrak{p}) = (\rho_{V \cap U_i \cap U_j} \circ \varphi)_{f(\mathfrak{p})}(s(f(\mathfrak{p}))) = f_j^\sharp(V \cap U_i \cap U_j)(s)(\mathfrak{p})$$

for all $\mathfrak{p} \in f^{-1}(V \cap U_i \cap U_j)$, so the sections agree on their overlap. Then they glue to a unique section $f^\sharp(V)(s) \in f_*\mathcal{O}_X(V)$.

It suffices to verify that α and β compose to the identity. Fix $(f, f^\sharp) \in \mathrm{Hom}_{\mathfrak{Sch}}(X, \mathrm{Spec} A)$, and denote $(\beta \circ \alpha)(f, f^\sharp) = (g, g^\sharp)$. Then $g|_{U_i} = g_i$, where

$$(g_i, g_i^\sharp) = \beta(\rho_{U_i} \circ f^\sharp(\mathrm{Spec} A)) = \beta(f^\sharp(U_i)).$$

This gives us $g_i = f|_{U_i}$ by the affine case, so that $g = f$.

Fix $V \subset \operatorname{Spec} A$ open and $s \in \mathcal{O}_{\operatorname{Spec} A}(V)$. Note that the $g^\sharp(V)(s)$ is the gluing of the sections $g^\sharp(V \cap U_i)(s)$ defined by

$$\begin{aligned} g^\sharp(V \cap U_i)(s)(\mathfrak{p}) &= (\rho_{V \cap U_i} \circ f^\sharp(V))_{g(\mathfrak{p})}(s(g(\mathfrak{p}))) \\ &= f^\sharp_{f(\mathfrak{p})}(s(f(\mathfrak{p}))) \\ &= f^\sharp(V \cap U_i)(s)(\mathfrak{p}) \end{aligned}$$

for all $\mathfrak{p} \in f^{-1}(V \cap U_i)$. Thus, $(f, f^\sharp) = (g, g^\sharp)$.

Finally, fix $\varphi : A \rightarrow \Gamma(X, \mathcal{O}_X)$ and let $(f, f^\sharp) = \beta(\varphi)$, $\psi = (\alpha)(f, f^\sharp)$. Then

$$\rho_{U_i} \circ \psi = \rho_{U_i} \circ (\alpha(f, f^\sharp)) = f^\sharp(U_i) = \rho_{U_i} \circ \varphi$$

by the affine case, so that $\psi = \varphi$ by uniqueness of gluing. We conclude that α is a bijection. $\square$

2.5. Describe $\operatorname{Spec} \mathbb{Z}$, and show that it is a final object for the category of schemes, i.e., each scheme X admits a unique morphism to $\operatorname{Spec} \mathbb{Z}$.

Proof. The only primes of $\mathbb{Z}$ are (p) for p prime, and (0) . The former are all maximal, while the latter is a generic point for $\operatorname{Spec} \mathbb{Z}$, since $V(\mathfrak{a}) \ni (0)$ implies $(0) \supset \mathfrak{a}$, i.e. $\mathfrak{a} = (0) \subset \mathfrak{p}$ for all $\mathfrak{p}$ prime.

We have a bijection

$$\operatorname{Hom}_{\mathbf{Sch}}(X, \operatorname{Spec} \mathbb{Z}) \rightarrow \operatorname{Hom}_{\mathbf{Ring}}(\mathbb{Z}, \Gamma(X, \mathcal{O}_X)),$$

so that the former is singleton for all X by the fact that $\mathbb{Z}$ is initial in $\mathbf{Ring}$. $\square$

2.7. Let X be a scheme. For any point $x \in X$, let $\mathcal{O}_x$ be the local ring at x , and m_x its maximal ideal. We define the *residue field* of x on X to be the field $k(x) = \mathcal{O}_x/m_x$. Now, let K be any field. Show that to give a morphism of $\operatorname{Spec} K$ to X , it is equivalent to giving a point $x \in X$ and an inclusion map $k(x) \hookrightarrow K$.

Proof. Note that $\operatorname{Spec} K = \{(0)\}$ is singleton, so a morphism $f : \operatorname{Spec} K \rightarrow X$ is a point $x = f((0)) \in X$, as well as the information $f^\sharp(U) : \mathcal{O}_X(U) \rightarrow K$ for every $x \in U \subset X$ open. Restriction is the identity on $\mathcal{O}_{\operatorname{Spec} K}$, so this defines a map $f_x^\sharp : \mathcal{O}_x \rightarrow K$. Since each element of m_x is not invertible, this gives us a map $\overline{f_x^\sharp} : k(x) \rightarrow K$. It is a morphism of fields, and thus an injection.

Conversely, suppose we have an inclusion $k(x) \hookrightarrow K$. Then, for each $x \in U \subset X$, we have a map

$$f^\sharp(U) : \mathcal{O}(U) \rightarrow \mathcal{O}_x \rightarrow k(x) \hookrightarrow K.$$

These maps play well with restriction, thus giving us a morphism $f^\sharp : \mathcal{O}_X \rightarrow \mathcal{O}_{\operatorname{Spec} K}$ of sheaves. Defining $f : \operatorname{Spec} K \rightarrow X : (0) \mapsto x$ gives us a morphism $(f, f^\sharp) : \operatorname{Spec} K \rightarrow X$ of schemes, as desired. $\square$

2.9. If X is a topological space, and Z an irreducible closed subset of X , a generic point for Z is a point ζ such that $Z = \{\zeta\}^-$. If X is a scheme, show that every (nonempty) irreducible closed subset has a unique generic point.

Proof. We begin by proving the case of $X = \operatorname{Spec} A$ affine. Let $Z = V(\mathfrak{q}) \subset X$ be irreducible and closed. That is, $V(\mathfrak{q}) = V(\mathfrak{a}) \cup V(\mathfrak{b}) = V(\mathfrak{ab})$ implies $V(\mathfrak{a}) = V(\mathfrak{q})$ or $V(\mathfrak{b}) = V(\mathfrak{q})$. Suppose now that $ab \in \mathfrak{q}$. Then $V((\mathfrak{q}, a)(\mathfrak{q}, b)) = V(\mathfrak{q})$ gives us $(\mathfrak{q}, a) \subset \sqrt{(\mathfrak{q}, a)} = \mathfrak{q}$ or $(\mathfrak{q}, b) \subset \sqrt{(\mathfrak{q}, b)} = \mathfrak{q}$. Hence, either $a \in \mathfrak{q}$ or $b \in \mathfrak{q}$, so that $\mathfrak{q}$ is prime.

Suppose that ζ is a generic point associated to $Z = V(\mathfrak{q})$. Then

$$\{\zeta\}^- = \bigcap_{V(\mathfrak{a}) \ni \zeta} V(\mathfrak{a}) = V(\cup_{\mathfrak{a} \subset \zeta} \mathfrak{a}) = V(\zeta)$$

implies that $\zeta = \mathfrak{q}$. This is clearly unique, and we showed that $\mathfrak{q}$ is prime above, so this is always possible.

Let Z be an irreducible closed subset of a scheme X , and let $\{U_i \cong \text{Spec } A_i\}_{i \in I}$ be an affine open cover of X . Replace $\{U_i\}$ by a subcover such that each $U_i \cap Z \neq \emptyset$. We claim that each $U_i \cap Z$ is irreducible in U_i . Take $\emptyset \neq U_i \cap V \subset U_i \cap Z$ nonempty open in U_i . Then $\emptyset \neq U_i \cap V \subset Z$ nonempty open gives us that

$$\text{Cl}_{U_i}(U_i \cap V) = U_i \cap \text{Cl}_X(U_i \cap V) = U_i \cap Z,$$

whence $U_i \cap Z$ is irreducible in U_i .

Take $\zeta_i \in U_i$ to be the unique generic point of $U_i \cap Z$ given by the affine case, so $\{\zeta_i\}^- \supset \text{Cl}_{U_i}(\zeta_i) = Z \cap U_i$. Note that $Z \cap U_i \cap U_j \neq \emptyset$ for all i, j , as otherwise

$$Z = Z \cap (U_i^c \cup U_j^c) = (Z \cap U_i^c) \cup (Z \cap U_j^c)$$

gives us that Z is disjoint from U_i or U_j . Then

$$\{\zeta_i\}^- \cap U_j \supset (Z \cap U_i) \cap U_j \neq \emptyset$$

implies $\zeta_i \in U_j$, as otherwise $U_j^c \cap \{\zeta_i\}^- \supset \{\zeta_i\}^-$. Then $\text{Cl}_{U_j}(\zeta_i) = \{\zeta_i\}^- \cap U_j$ implies

$$\text{Cl}_{U_j}(\zeta_i) \supset (Z \cap U_i \cap U_j)^- \cap U_j = Z \cap U_j.$$

Hence, ζ_i is a generic point of $Z \cap U_j \subset U_j$ for each j . Uniqueness in the affine case gives us $\zeta_i = \zeta_j =: \zeta$ for all i, j , so that

$$\{\zeta\}^- \supset \bigcup_{i \in I} U_i \cap Z = Z.$$

Any generic point ζ' of Z must also be a generic point of each of the $U_i \cap Z$, so uniqueness follows from the affine case. $\square$